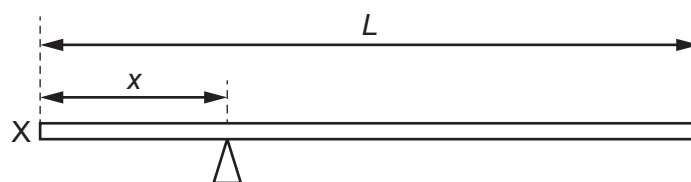


- 14 The diagram shows a uniform bar of mass M and length L resting on a pivot at a distance x from end X.



An object of mass m is placed on the bar at distance y from the pivot so that the bar is in equilibrium.

What is an expression for y ?

- A $\frac{xM}{m}$ B $\frac{M}{m}(L - x)$ C $\frac{M}{2m}(L - 2x)$ D $\frac{1}{m}(Lm - xM)$

- 15 The mass and volume of an object are varied.

Which two changes, when made together, **must** increase the density of the object?

	mass	volume
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

- 16 On Earth, a solid object that is fully submerged in a liquid experiences an upthrust U_E .

On Mars, the same object, fully submerged in the same liquid, experiences an upthrust U_M .

The acceleration of free fall on Mars is 3.7 m s^{-2} .

Assume that the liquid's density and the object's volume have the same values on Earth and Mars.

What is the ratio $\frac{U_M}{U_E}$?

- A 0.38 B 1.0 C 2.7 D 3.6

- 17 A student attempts to derive the formula for kinetic energy E_K . She begins by considering an object of mass m that is initially at rest. A constant force F applied to the object causes it to accelerate to final velocity v in displacement s . The kinetic energy gained by the object is equal to the work done on the object by the force F .

Which equation does the student **not** need in order to derive the formula for E_K ?

- A $F = ma$ B $W = Fs$ C $E = \frac{1}{2}Fs$ D $v^2 = u^2 + 2as$